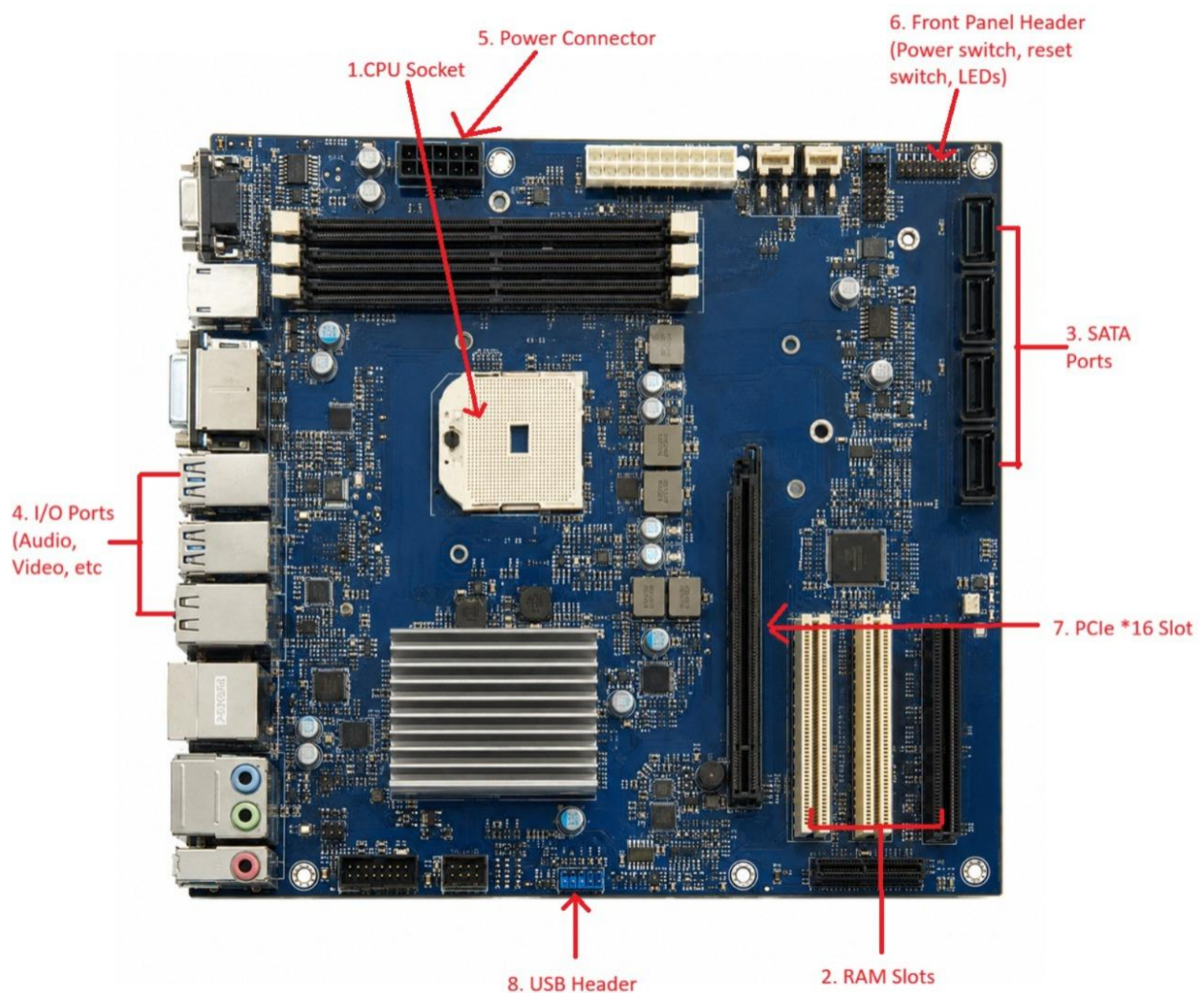


Assignment - 1.2

Day 2 : Motherboard and CPU theory : Types of motherboards and CPU socket Compatibility. Functions of the CPU and chipset. Practical: Identify the motherboard slots and ports.

Task – 1

Draw a labelled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



» Names of the Motherboards slots and ports :-

1. **CPU Socket** – The CPU is placed here.
2. **RAM Slots** – Used to install RAM.
3. **SATA Ports** – Used to connect HDDs or SSDs.
4. **I/O Ports** – Used to connect external devices like USB, audio and video devices.
5. **Power Connector** – Supplies power to the motherboard/CPU.
6. **Front Panel Header** – Connects the power button, reset button and LEDs.
7. **PCIe x16 Slot** – Used to connect a graphics card.
8. **USB Header** – Connects the computer case's front USB ports.

Task – 2

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. LGA1151 :-

LGA1151 is a CPU socket developed by Intel. It is mainly used with Intel 6th, 7th, 8th, and 9th generation Core processors, depending on the motherboard chipset. The socket has **1,151 contact points**.

Compatible Model for LGA1151:- Intel Core i7-9700K — a 9th-generation desktop processor that uses the LGA1151 socket.

2. AM4 :-

AM4 is a CPU socket developed by AMD. It was widely used for AMD Ryzen processors and supports several generations of Ryzen CPUs. AM4 motherboards commonly support features such as DDR4 RAM and PCIe connectivity.

Compatible Model for AM4 :- AMD Ryzen 5 5600 — a popular 6-core, 12-thread processor that uses the AM4 socket.

3. LGA1200 :-

LGA1200 is an Intel CPU socket designed for 10th and 11th generation Intel desktop processors. As its name suggests, it has **1,200 contact points** between the CPU and motherboard.

Compatible Model for LGA1200 :- Intel Core i5-10600K — a 10th-generation 6-core processor designed for the LGA1200

Task – 3

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit):-

- » The CPU is the main processor or “brain” of the computer. It executes instructions, processes data, and controls the tasks performed by different programs. When you open an application, the CPU processes the instructions required to start and run that application.

Chipset :-

- » The chipset is a group of controllers on the motherboard that helps the CPU communicate with other hardware components. It manages communication between devices such as USB ports, storage devices, PCIe devices, and other motherboard components.

What happened when we try to open SPOTIFY ? :-

1. You click the Spotify icon, and the computer sends the instruction to start the application.
2. The CPU processes these instructions and begins running Spotify.
3. The chipset helps coordinate communication between the CPU and other hardware, such as the storage drive and RAM.
4. Spotify's required files are transferred from storage into RAM, with the CPU processing the instructions needed to launch the app.
5. Once Spotify opens, the CPU continuously processes commands, while the chipset helps different hardware components communicate smoothly.
6. When you play a song, the CPU processes the app's instructions and the audio data is sent through the computer's audio hardware so you can hear the music.

Task – 4

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

